Cambridge Assessment International Education

# Cambridge O Level

**CHEMISTRY** **5070/12**

Paper 1 Multiple Choice **May/June 2025**

**1 hour**

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

**INSTRUCTIONS**

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

**INFORMATION**

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.


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**[Turn over**

**1** At temperature X, the volume of samples of hexane and pentane are **not** affected by changing the pressure from 1 atm to 2 atm.

At temperature X, the pentane molecules are free to move.

At temperature X, the hexane molecules are in fixed positions.

What are the states of hexane and pentane at temperature X?

| | hexane | pentane |
|---|---|---|
| **A** | liquid | solid |
| **B** | liquid | gas |
| **C** | solid | liquid |
| **D** | solid | gas |

**2** A gas is produced by a chemical reaction at one side of a laboratory.

After a few minutes, the gas is detected at the other side of the laboratory.

Which process explains why the gas is detected at the other side of the laboratory?

**A** condensation

**B** diffusion

**C** dissolving

**D** evaporation

**3** Which statement is correct?

**A** All compounds are ionic.

**B** All compounds conduct electricity when molten.

**C** Each atom of an element contains the same number of protons.

**D** In a mixture of substances, the proportions of the substances are always the same.

**4** Atoms of element Q form positive ions.

An ion of Q has 10 electrons and 14 neutrons.

Which statement is correct?

**A** An atom of Q has only two occupied electron shells.

**B** Q is a non-metal.

**C** The atomic number of Q is 10.

**D** The mass number of Q is greater than 24.

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**5** Which particle contains the greatest number of electrons?

**A** $Mg^{2+}$ **B** $N^{3-}$ **C** Ne **D** $S^{2-}$

**6** The table shows data about the two isotopes in a sample of the element europium.

| relative mass of isotope | percentage abundance of isotope |
|---|---|
| 150.92 | 47.81 |
| 152.92 | 52.19 |

What is the relative atomic mass of this sample of europium?

**A** 151.88 **B** 151.96 **C** 152.00 **D** 152.04

**7** Which statement about magnesium oxide is correct?

**A** It has a high melting point and good electrical conductivity at r.t.p.

**B** It is made up of magnesium anions and oxygen cations.

**C** It is an ionic compound and has good electrical conductivity when molten.

**D** It is strongly bonded because of electrons shared between magnesium atoms and oxygen atoms.

**8** Which statement about silicon(IV) oxide is correct?

**A** It conducts electricity.

**B** It has a giant covalent structure.

**C** It is an alloy.

**D** It is ionically bonded.

**9** Which compound has the highest relative formula mass, $M_r$?

[$A_r$: K, 39; N, 14; O, 16; Ca, 40; C, 12; Li, 7; S, 32; Mg, 24; C*l*, 35.5]

**A** calcium carbonate

**B** lithium sulfate

**C** magnesium chloride

**D** potassium nitrate

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**10** 100 $cm^3$ of ethene is completely combusted in 400 $cm^3$ of oxygen. The volumes of both gases are measured at r.t.p.

What is the final volume of gas in the mixture measured at r.t.p.?

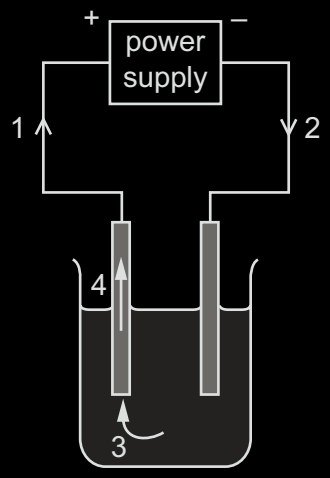
**A** 200 $cm^3$ **B** 250 $cm^3$ **C** 300 $cm^3$ **D** 500 $cm^3$

**11** The formula of hydrated sodium carbonate is $Na_2CO_3 \cdot 10H_2O$.

What is the percentage composition by mass of oxygen in hydrated sodium carbonate?

**A** 16.8 **B** 22.4 **C** 55.9 **D** 72.7

**12** The diagram shows the electrolysis of molten lead bromide with inert electrodes.




Which row identifies the particles that carry charge along each arrow?

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **A** | bromide ions | electrons | lead ions | electrons |
| **B** | electrons | electrons | bromide ions | lead ions |
| **C** | electrons | electrons | bromide ions | electrons |
| **D** | electrons | electrons | electrons | electrons |

**13** What are the products formed at the two electrodes when aqueous copper(II) sulfate is electrolysed using copper electrodes?

| | anode product | cathode product |
|---|---|---|
| **A** | aqueous copper(II) ions | copper metal |
| **B** | aqueous copper(II) ions | hydrogen gas |
| **C** | oxygen gas | copper metal |
| **D** | oxygen gas | hydrogen gas |

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**14** Which statement about a hydrogen–oxygen fuel cell is correct?

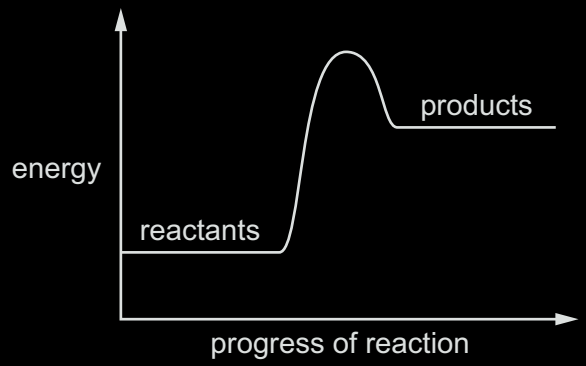
**A** It uses a chemical reaction that has only one product.

**B** It works by burning hydrogen in air.

**C** It requires a continuous input of electrical energy.

**D** It is less efficient than a petrol engine.

**15** A reaction pathway diagram is shown.




Which equation does the reaction pathway diagram show?

**A** $2H(g) + O(g) \rightarrow H_2O(g)$

**B** $H_2(g) \rightarrow 2H(g)$

**C** $H_2O(g) \rightarrow H_2O(l)$

**D** $2O(g) \rightarrow O_2(g)$

**16** Which process is a chemical change?

**A** the distillation of aqueous ethanol

**B** the evaporation of water from aqueous sodium chloride

**C** the melting of wax

**D** the rusting of iron

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**[Turn over**

**17** Aqueous sodium thiosulfate reacts with hydrochloric acid. The rate of the reaction increases if the concentration of both reactants is increased.

Nitrogen gas reacts with hydrogen gas. The rate of the reaction increases if the pressure in the reaction vessel is increased.

Which row correctly explains why the given change increases the rate of the reaction?

| | increasing the concentration of aqueous sodium thiosulfate and hydrochloric acid | increasing the pressure in the reaction vessel containing nitrogen and hydrogen |
|---|---|---|
| **A** | higher frequency of collisions between particles | higher frequency of collisions between particles |
| **B** | higher frequency of collisions between particles | the activation energy is decreased |
| **C** | the activation energy is decreased | higher frequency of collisions between particles |
| **D** | the activation energy is decreased | the activation energy is decreased |

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**18** Magnesium reacts with dilute sulfuric acid.

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

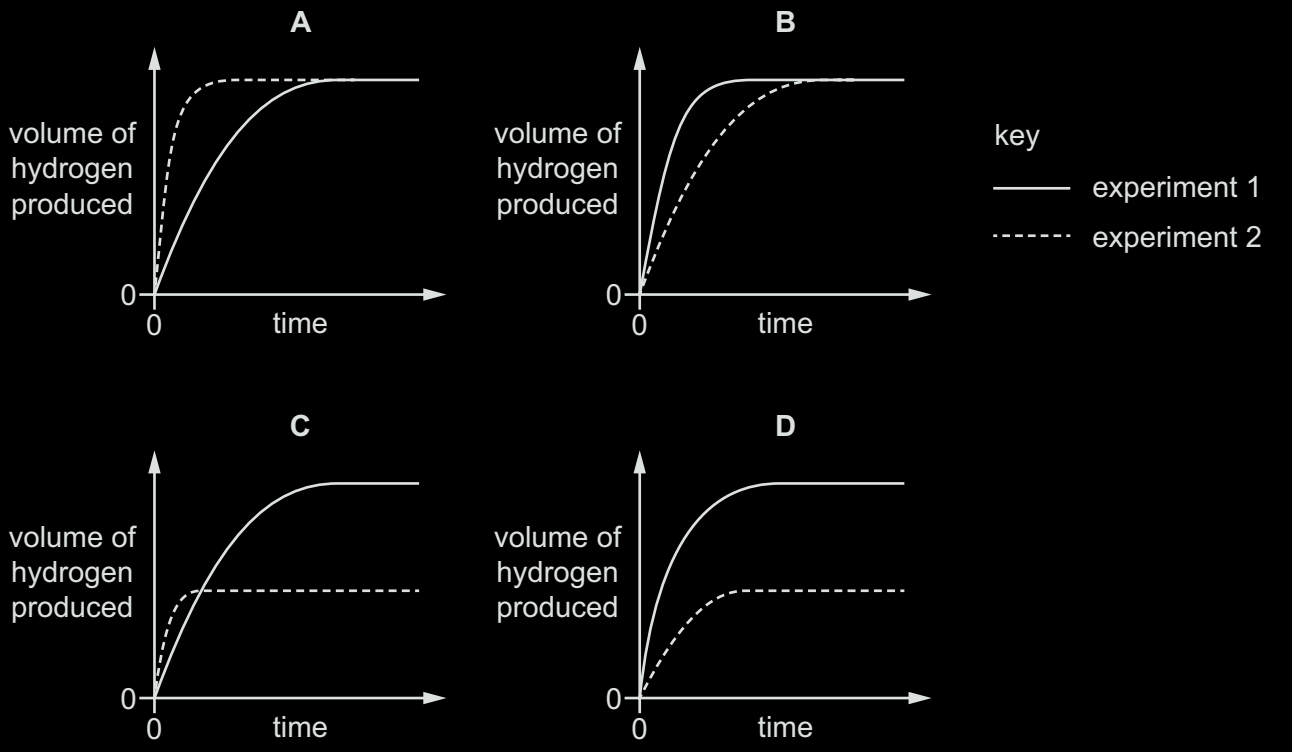
Two experiments are carried out at 25 °C.

experiment 1 24.0 g of powdered magnesium is reacted with 100 $cm^3$ of 1.0 $mol/dm^3$ sulfuric acid.

experiment 2 24.0 g of powdered magnesium is reacted with 50 $cm^3$ of 2.0 $mol/dm^3$ sulfuric acid.

During each experiment, the volume of hydrogen produced is measured at regular time intervals. The results are plotted on a graph.

Which graph is correct?

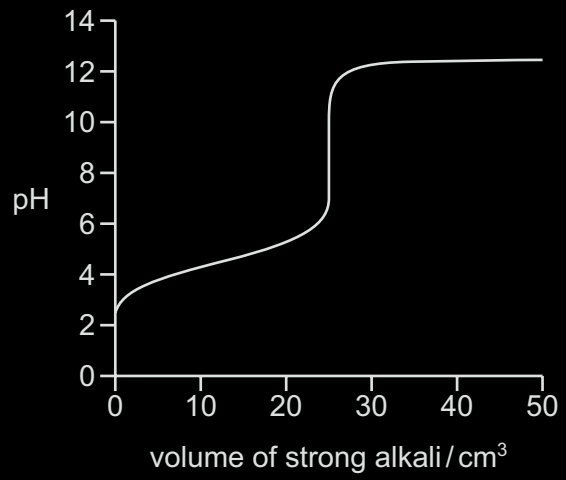



**19** Which row gives the catalyst for each of the two processes?

| | Contact process | Haber process |
|---|---|---|
| **A** | iron | iron |
| **B** | iron | nickel |
| **C** | vanadium(V) oxide | iron |
| **D** | vanadium(V) oxide | nickel |

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**20** The diagram shows the change in pH as a strong alkali is added to a weak acid.




40 $cm^3$ of a strong alkali is added to a weak acid and a sample of the solution is tested separately with methyl orange and thymolphthalein.

Which row is correct?

| | methyl orange test | thymolphthalein test |
|---|---|---|
| **A** | red | colourless |
| **B** | yellow | colourless |
| **C** | red | blue |
| **D** | yellow | blue |

**21** Which row shows the equations for the dissociation of methanoic acid and of nitric acid?

| | methanoic acid | nitric acid |
|---|---|---|
| **A** | $HCOOH(aq) \rightarrow H^+(aq) + HCOO^-(aq)$ | $HNO_3(aq) \rightarrow H^+(aq) + NO_3^-(aq)$ |
| **B** | $HCOOH(aq) \rightarrow H^+(aq) + HCOO^-(aq)$ | $HNO_3(aq) \rightleftharpoons H^+(aq) + NO_3^-(aq)$ |
| **C** | $HCOOH(aq) \rightleftharpoons H^+(aq) + HCOO^-(aq)$ | $HNO_3(aq) \rightleftharpoons H^+(aq) + NO_3^-(aq)$ |
| **D** | $HCOOH(aq) \rightleftharpoons H^+(aq) + HCOO^-(aq)$ | $HNO_3(aq) \rightarrow H^+(aq) + NO_3^-(aq)$ |

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**22** Which statement about elements in the Periodic Table is correct?

**A** Across a period, elements at the left-hand side of the Periodic Table are more metallic than those at the right-hand side.

**B** Down a group, elements at the top of the group lose electrons more readily than those at the bottom of the group.

**C** Elements in the same group of the Periodic Table have the same number of completed shells of electrons.

**D** Elements in the same period of the Periodic Table have the same number of electrons in the outer shell.

**23** Rubidium and caesium are both elements in Group I of the Periodic Table.

Which prediction comparing rubidium and caesium is correct?

**A** Caesium has the higher density and the greater reactivity.

**B** Caesium has the higher density; rubidium has the greater reactivity.

**C** Rubidium has the higher density and the greater reactivity.

**D** Rubidium has the higher density; caesium has the greater reactivity.

**24** Three statements about the halogens are listed.

1 All halogens are non-metallic diatomic molecules.

2 Chlorine displaces both bromine and iodine from aqueous solutions of their salts.

3 The halogens become more reactive on descending Group VII of the Periodic Table.

Which statements are correct?

**A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

**25** Which statements about neon are correct?

1 It is unreactive.

2 It has strong intermolecular forces.

3 It has a full outer shell of electrons.

**A** 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 3 only

**26** Substance X has the following properties.

- melting point 1237 °C
- boiling point 2595 °C
- malleable
- good electrical and thermal conductivity at r.t.p.

What is the structure of substance X?

**A** giant covalent

**B** giant ionic lattice

**C** giant metallic lattice

**D** simple molecular

**27** Which statement explains why objects manufactured from aluminium corrode very slowly?

**A** Aluminium is above hydrogen in the reactivity series.

**B** Aluminium is below hydrogen in the reactivity series.

**C** The objects become coated with a protective layer of aluminium nitride.

**D** The objects become coated with a protective layer of aluminium oxide.

**28** The iron(III) oxide used in a blast furnace contains the impurity silicon(IV) oxide, $SiO_2$.

Which reaction removes the silicon(IV) oxide?

**A** $3SiO_2 + Fe_2O_3 \rightarrow 2Fe + 3SiO_3$

**B** $SiO_2 + C \rightarrow Si + CO_2$

**C** $SiO_2 + CaO \rightarrow CaSiO_3$

**D** $2SiO_2 + N_2 \rightarrow 2Si + 2NO_2$

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**29** The domestic water supply is treated before it is supplied to houses. Three treatments are listed.

- sedimentation and filtration
- treatment with chlorine
- treatment with carbon

Which row shows the purpose of the treatments?

| | sedimentation and filtration | treatment with chlorine | treatment with carbon |
|---|---|---|---|
| **A** | kills microbes | removes solids | removes unpleasant tastes and odours |
| **B** | removes solids | kills microbes | removes unpleasant tastes and odours |
| **C** | removes solids | removes unpleasant tastes and odours | kills microbes |
| **D** | removes unpleasant tastes and odours | kills microbes | removes solids |

**30** Nitrates and ammonium salts are used as fertilisers.

Which fertiliser contains the greatest mass of nitrogen in 50 kg of the compound?

[$A_r$: H, 1; N, 14; O, 16; S, 32; C*l*, 35.5; K, 39]

**A** $NH_4Cl$ **B** $NH_4NO_3$ **C** $(NH_4)_2SO_4$ **D** $KNO_3$

**31** Which strategy reduces methane emissions?

**A** flue gas desulfurisation

**B** planting more trees

**C** reduction in livestock farming

**D** use of catalytic converters

**32** Compound P is an alcohol with the molecular formula $C_4H_{10}O$.

How many structural isomers are there of compound P?

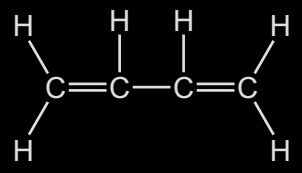
**A** 2 **B** 3 **C** 4 **D** 5

**33** Alkanes undergo a substitution reaction with chlorine.

Which row shows two correct statements about the reaction between alkanes and chlorine?

| | statement 1 | statement 2 |
|---|---|---|
| **A** | a carbon atom is replaced by a chlorine atom | the reaction requires thermal energy |
| **B** | a hydrogen atom is replaced by a chlorine atom | the reaction requires thermal energy |
| **C** | a carbon atom is replaced by a chlorine atom | the reaction requires ultraviolet light |
| **D** | a hydrogen atom is replaced by a chlorine atom | the reaction requires ultraviolet light |

**34** A molecule of the compound $C_4H_6$ is shown.




This molecule undergoes two separate addition reactions. It undergoes an addition reaction with excess bromine and an addition reaction with steam.

One molecule of $C_4H_6$ reacts with .....1..... of bromine.

When $C_4H_6$ reacts with steam, .....2..... is formed.

Which words complete gaps 1 and 2?

| | 1 | 2 |
|---|---|---|
| **A** | one molecule | an alcohol |
| **B** | one molecule | a carboxylic acid |
| **C** | two molecules | an alcohol |
| **D** | two molecules | a carboxylic acid |

**35** Which statement about members of the homologous series of alcohols is correct?

**A** An alcohol with two carbon atoms in each molecule is called methanol.

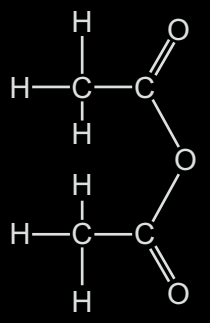
**B** Butanol can be combusted to give carbon dioxide and water only.

**C** Ethanol is the only alcohol that can be oxidised to a carboxylic acid.

**D** Propanol can be made by the catalysed addition of steam to ethene.

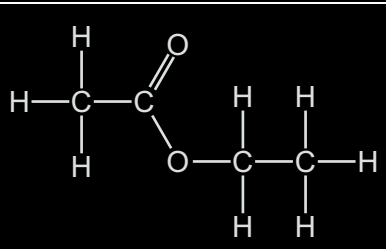
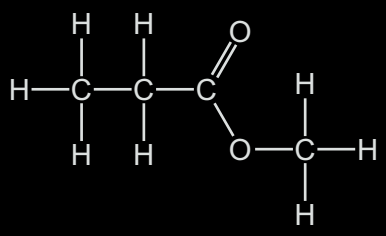
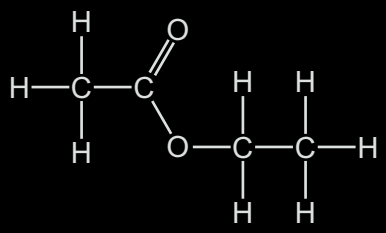
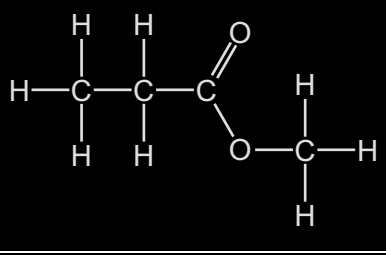
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**36** The diagram shows the structure of a compound called ethanoic anhydride.



1 mol of ethanoic anhydride reacts with water to form 2 mol of a carboxylic acid only. This carboxylic acid reacts with ethanol to form an ester.

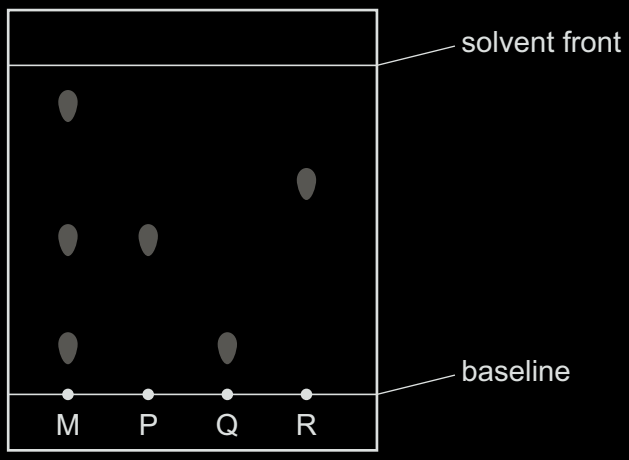
How many moles of water react with 1 mol of the ethanoic anhydride and what is the structure of the ester?

| | number of moles of water | structure of the ester |
|---|---|---|
| **A** | 1 |  |
| **B** | 1 |  |
| **C** | 2 |  |
| **D** | 2 |  |

**37** Four samples are analysed using paper chromatography.

P, Q and R are pure substances.

M is a mixture of different substances.




Which substances does mixture M contain?

**A** P, Q and R

**B** P, Q and another substance

**C** P and Q only

**D** P and R only

**38** A student is making a sample of aqueous copper(II) sulfate. The student adds an excess of copper(II) oxide powder to warm sulfuric acid and stirs the mixture.

Which apparatus is used to separate aqueous copper(II) sulfate from the excess copper(II) oxide?

**A** burette

**B** distillation apparatus

**C** filter funnel and paper

**D** measuring cylinder

**39** How can a pure sample of barium sulfate be obtained from barium carbonate?

**A** Dissolve it in dilute hydrochloric acid, add dilute sulfuric acid, filter and crystallise.

**B** Dissolve it in dilute hydrochloric acid, add dilute sulfuric acid, filter and wash.

**C** Dissolve it in water, add dilute sulfuric acid, filter and crystallise.

**D** Dissolve it in water, add dilute sulfuric acid, filter and wash.

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**40** A sample of a white powder, X, is dissolved in water.

Tests are done on separate portions of this solution and the observations are shown in the table.

| test | observation |
|---|---|
| add aqueous sodium hydroxide a drop at a time until in excess | white precipitate forms, soluble in excess, giving a colourless solution |
| acidify with dilute nitric acid then add aqueous silver nitrate | white precipitate forms |
| acidify with dilute nitric acid then add aqueous barium nitrate | the solution remains colourless |

What is X?

**A** aluminium chloride

**B** aluminium sulfate

**C** calcium chloride

**D** calcium sulfate

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## The Periodic Table of Elements

| Group I | Group II | | | | | | | | | | | Group III | Group IV | Group V | Group VI | Group VII | Group VIII |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 1 H hydrogen 1 | | | | | | | | | | 2 He helium 4 |
| 3 Li lithium 7 | 4 Be beryllium 9 | | | | | | | | | | | 5 B boron 11 | 6 C carbon 12 | 7 N nitrogen 14 | 8 O oxygen 16 | 9 F fluorine 19 | 10 Ne neon 20 |
| 11 Na sodium 23 | 12 Mg magnesium 24 | | | | | | | | | | | 13 Al aluminium 27 | 14 Si silicon 28 | 15 P phosphorus 31 | 16 S sulfur 32 | 17 Cl chlorine 35.5 | 18 Ar argon 40 |
| 19 K potassium 39 | 20 Ca calcium 40 | 21 Sc scandium 45 | 22 Ti titanium 48 | 23 V vanadium 51 | 24 Cr chromium 52 | 25 Mn manganese 55 | 26 Fe iron 56 | 27 Co cobalt 59 | 28 Ni nickel 59 | 29 Cu copper 64 | 30 Zn zinc 65 | 31 Ga gallium 70 | 32 Ge germanium 73 | 33 As arsenic 75 | 34 Se selenium 79 | 35 Br bromine 80 | 36 Kr krypton 84 |
| 37 Rb rubidium 85 | 38 Sr strontium 88 | 39 Y yttrium 89 | 40 Zr zirconium 91 | 41 Nb niobium 93 | 42 Mo molybdenum 96 | 43 Tc technetium – | 44 Ru ruthenium 101 | 45 Rh rhodium 103 | 46 Pd palladium 106 | 47 Ag silver 108 | 48 Cd cadmium 112 | 49 In indium 115 | 50 Sn tin 119 | 51 Sb antimony 122 | 52 Te tellurium 128 | 53 I iodine 127 | 54 Xe xenon 131 |
| 55 Cs caesium 133 | 56 Ba barium 137 | 57–71 lanthanoids | 72 Hf hafnium 178 | 73 Ta tantalum 181 | 74 W tungsten 184 | 75 Re rhenium 186 | 76 Os osmium 190 | 77 Ir iridium 192 | 78 Pt platinum 195 | 79 Au gold 197 | 80 Hg mercury 201 | 81 Tl thallium 204 | 82 Pb lead 207 | 83 Bi bismuth 209 | 84 Po polonium – | 85 At astatine – | 86 Rn radon – |
| 87 Fr francium – | 88 Ra radium – | 89–103 actinoids | 104 Rf rutherfordium – | 105 Db dubnium – | 106 Sg seaborgium – | 107 Bh bohrium – | 108 Hs hassium – | 109 Mt meitnerium – | 110 Ds darmstadtium – | 111 Rg roentgenium – | 112 Cn copernicium – | 113 Nh nihonium – | 114 Fl flerovium – | 115 Mc moscovium – | 116 Lv livermorium – | 117 Ts tennessine – | 118 Og oganesson – |

**Key**

atomic number
atomic symbol
name
relative atomic mass

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| lanthanoids | 57 La lanthanum 139 | 58 Ce cerium 140 | 59 Pr praseodymium 141 | 60 Nd neodymium 144 | 61 Pm promethium – | 62 Sm samarium 150 | 63 Eu europium 152 | 64 Gd gadolinium 157 | 65 Tb terbium 159 | 66 Dy dysprosium 163 | 67 Ho holmium 165 | 68 Er erbium 167 | 69 Tm thulium 169 | 70 Yb ytterbium 173 | 71 Lu lutetium 175 |
| actinoids | 89 Ac actinium – | 90 Th thorium 232 | 91 Pa protactinium 231 | 92 U uranium 238 | 93 Np neptunium – | 94 Pu plutonium – | 95 Am americium – | 96 Cm curium – | 97 Bk berkelium – | 98 Cf californium – | 99 Es einsteinium – | 100 Fm fermium – | 101 Md mendelevium – | 102 No nobelium – | 103 Lr lawrencium – |

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).